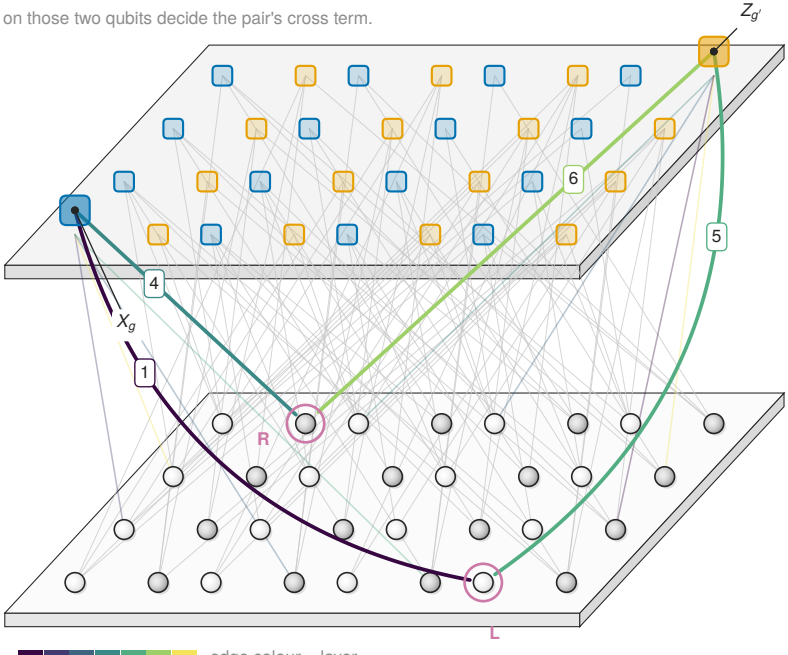
checks above, data qubits below

X_g and $Z_{g'}$ share one L and one R qubit; only the four CNOTs on those two qubits decide the pair's cross term.



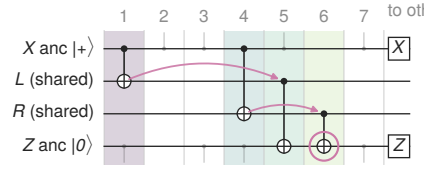
edge colour = layer

1 2 3 4 5 6 7

columns = the 7 (6) CNOT layers of the whole circuit; only gates on these four wires are shown

same pair, published cycle (depth 7): proper

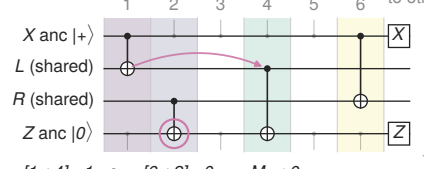
wires L, R are the two ringed qubits at left



$o_L = [1 < 5] = 1$, $o_R = [4 < 6] = 1$ → proper

y enters Z-anc twice (via L, via R): $y + y = 0$ → proper

same pair, Kőnig colouring (depth 6): not proper



$o_L = [1 < 4] = 1$, $o_R = [6 < 2] = 0$ → $M_\pi \neq 0$

y enters once: outcome $+y$ → $M_\pi \neq 0$

degree bound $\Delta = 6$ · Kőnig colouring: depth 6, $M_\pi \neq 0$ · published cycle: depth 7, proper · **this paper: 7 is optimal**